Binary to Octal Conversions

To convert a binary number to octal, we can first convert it to decimal and then convert the decimal number to octal. However, there is a much easier method.

- Starting from the right, divide the binary number into groups of three digits (bits).
- If the leftmost group has fewer than three bits, add leading zeros.
- Convert each group of three bits to its decimal equivalent (0 through 7).
- Write the resulting digits in the same order to obtain the octal number.

Example

Convert 1101011_2 to octal.

Solution:

Group the digits from left to right.

001 101 011

Convert each group.

$$001 = 2^0 = 1$$
$$101 = 2^2 + 2^0 = 4 + 1 = 5$$
$$011 = 2^1 + 2^0 = 2 + 1 = 3$$

Therefore, $1101011_2 = 153_8$.

Convert the following binary numbers to octal form.

110010001_2

1010011111_2

11101011001_2

Converting Octal to Binary

To convert an octal number to binary, we reverse the process.

- Convert each octal digit to its three-bit binary equivalent.
- Combine the groups in the same order.
- Remove any unnecessary leading zeros.

Example

Convert 347_8 to binary.

Solution:

Convert each digit.

$$3 = 011$$
$$4 = 100$$
$$7 = 111$$

Combine the groups.

$$011\ 100\ 111$$

Remove the unnecessary leading zero.

$$347_8 = 11100111_2$$

Convert the following octal numbers to binary form.

621_8

4237_8

3052_8